LAB SESSION 11

Programming Exercise:

Question 1: Create a Book class that has title, author, and a Date object for the publication date. Implement a method to display the book details.

```cpp
#include <iostream>
using namespace std;

// Date class
class Date {
public:
    int day, month, year;

    // Constructor
    Date(int d, int m, int y) {
        day = d;
        month = m;
        year = y;
    }

    // Method to display date
    void displayDate() {
        cout << day << "/" << month << "/" << year;
    }
};

// Book class
class Book {
public:
    string title;
    string author;
    Date publicationDate;

    // Constructor
    Book(string t, string a, Date d) : publicationDate(d)
{       title = t;
        author = a;
    }

    // Method to display book details
    void displayDetails() {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Publication Date: ";
        publicationDate.displayDate();
        cout << endl;
    }
};

// Main function
int main() {
    // Create Date object
    Date pubDate(20, 5, 2025);

    // Create Book object
    Book myBook("C++ Basics", "John Doe", pubDate);

    // Display book details
    myBook.displayDetails();

    return 0;
}
```

```
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o
 tut1 } ; if ($?) { .\tut1 }
Title: C++ Basics
Author: John Doe
Publication Date: 20/5/2025
PS C:\C++ PROGRAMS>
```

Question 2: Create a Library class that can hold multiple Book objects. Implement methods to add books and display all books in the library.

```
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o
 tut1 } ; if ($?) { .\tut1 }
Books in Library:
Title: C++ Basics
Author: John Doe
Publication Date: 20/5/2025
------------------------
Title: OOP Concepts
Author: Jane Smith
Publication Date: 10/3/2024
------------------------
PS C:\C++ PROGRAMS>
```

```cpp
#include <iostream>
#include <vector>
using namespace std;

// Date class
class Date {
public:
    int day, month, year;

    Date(int d, int m, int y) {
        day = d;
        month = m;
        year = y;
    }

    void displayDate() {
        cout << day << "/" << month << "/" << year;
    }
};

// Book class
class Book {
public:
    string title;
    string author;
    Date publicationDate;

    Book(string t, string a, Date d) : publicationDate(d) {
        title = t;
        author = a;
    }

    void displayDetails() {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Publication Date: ";
        publicationDate.displayDate();
        cout << endl << "-------------------------" << endl;
    }
};

// Library class
class Library {
private:
    vector<Book> books;

public:
    // Add a book to the library
    void addBook(Book b) {
        books.push_back(b);
    }

    // Display all books in the library
    void displayAllBooks() {
        if (books.empty()) {
            cout << "Library is empty." << endl;
        } else {
            cout << "Books in Library:" << endl;
            for (Book b : books) {
                b.displayDetails();
            }
        }
    }
};

// Main function
int main() {
    // Create a library
    Library myLibrary;

    // Create some books
    Book book1("C++ Basics", "John Doe", Date(20, 5, 2025));
    Book book2("OOP Concepts", "Jane Smith", Date(10, 3,
2024));
    // Add books to library
    myLibrary.addBook(book1);
    myLibrary.addBook(book2);

    // Display all books
    myLibrary.displayAllBooks();

    return 0;
}
```